A16726W1

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B1. FLUIDS

TRINITY TERM 2023

Friday, 9th June, 9:30 am – 11.30 pm

Answer **two** *questions.*

Start the answer to each question in a **fresh book.**

The use of approved calculators is permitted.

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

Page 2 contains physics formulae and data for this paper.
The questions start on page 3.

The Navier–Stokes equation for viscous, incompressible fluid flow under gravity is

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla)\mathbf{v} + \frac{1}{\rho}\nabla p + g\hat{\mathbf{z}} = \nu \nabla^2 \mathbf{v} \ ,$$

where $\mathbf{v}$ is the fluid velocity, ρ the density, p the pressure, g the acceleration due to gravity, $\hat{\mathbf{z}}$ the vertical unit vector and ν the kinematic viscosity.

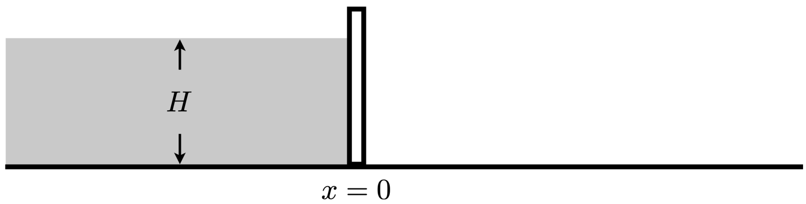
$1000\,\mathrm{mb} = 1000\,\mathrm{hPa} = 10^5\,\mathrm{Pa}$.

1. (a) Starting from the steady-state Euler equations for a fluid of uniform density under the influence of uniform gravitational acceleration, show that

$$\frac{\mathbf{v}\cdot\mathbf{v}}{2} + \frac{p}{\rho} + gz = \text{constant}$$

along any streamline, and provide a physical interpretation of this equation. [5]

(b) A large reservoir of water of depth H and cross-sectional width L is contained in the region $x < 0$ by a closed vertical sluice gate, as sketched in the figure.

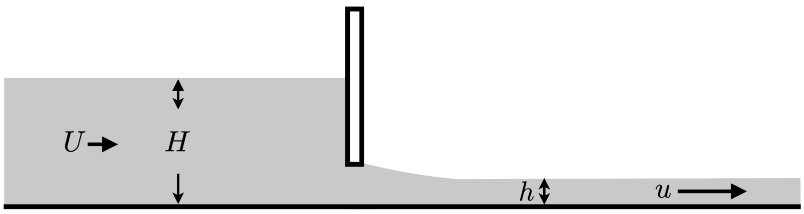


Show that the force exerted on the sluice gate is

$$\frac{\rho g H^2 L}{2},$$

where ρ is the density of water and g is the gravitational acceleration. [3]

(c) The sluice gate is now raised slightly to allow a flow to pass underneath. As sketched in the figure, the flow speed is U for $x \ll 0$ and u for $x \gg 0$, both independent of depth. The height of the water far downstream of the gate is h.



By applying the result from part (a) to an appropriate streamline, and also applying an additional constraint on the flow, show that $\hat{h} = h/H$ is given by either

$$\hat{h} = 1 \quad \text{or} \quad \hat{h} = \frac{F^2 + F\sqrt{F^2+8}}{4}$$

where $F = U/\sqrt{gH}$. [8]

(d) Write down the momentum budget for the fluid contained between a vertical section far upstream and a vertical section far downstream of the gate. By relating the fluxes of momentum at the vertical sections to the forces acting on the fluid between the sections, show that the force exerted on the gate is

$$\frac{\rho g H^2 L}{2}\,\frac{(1-\hat{h})^3}{(1+\hat{h})}.$$

[9]

2. (a) Describe what is meant by a *potential flow* and state the conditions under which such a flow is obtained. [3]

(b) Show that the real and imaginary parts of any complex analytic function, $w(z) = \phi(x,y) + i\psi(x,y)$ where $z = x + iy$, satisfy the Cauchy–Riemann relations:

$$\frac{\partial \phi}{\partial x} = \frac{\partial \psi}{\partial y}, \quad \frac{\partial \psi}{\partial x} = -\frac{\partial \phi}{\partial y}.$$

Explain how and why this result is useful for modelling two-dimensional potential flows. [9]

(c) Obtain an expression for the stream function in (r, θ) coordinates for the potential flow described by

$$w = A\sqrt{z}.$$

Sketch the stream lines for this flow and identify the physical flow that it represents. A paddle wheel is a floating object that moves with the flow, and rotates at a rate proportional to the vorticity. Describe, with the aid of a sketch, the movement of a paddle wheel introduced into this potential flow at finite r with θ very slightly less than 2π. [6]

(d) Define the *Reynolds number* and state its physical significance. State and give a physical explanation for two differences between the potential flow solution obtained in part (c) and the flow that would be obtained for the same situation in a real fluid at high Reynolds number. Sketch how the motion of the paddle wheel would differ if it were introduced into the flow at finite r and θ very slightly less than 2π. Why does the flow not tend towards the potential flow solution in the limit $\mathrm{Re} \to \infty$? [7]

3. (a) Consider a stably stratified, incompressible fluid with initial density $\rho = \rho_0(z)$. Show that a fluid parcel that is displaced vertically and then released will oscillate at the buoyancy frequency,

$$N = \sqrt{-\frac{g}{\rho_0}\frac{\mathrm{d}\rho_0}{\mathrm{d}z}}.$$

State any necessary assumptions. [4]

(b) Write down the Euler equation for linear perturbations about a state of rest in the x-z plane. Hence show that

$$\frac{\partial}{\partial t}\left(\frac{\partial v_x}{\partial z} - \frac{\partial v_z}{\partial x}\right) = -\frac{\partial b}{\partial x},$$

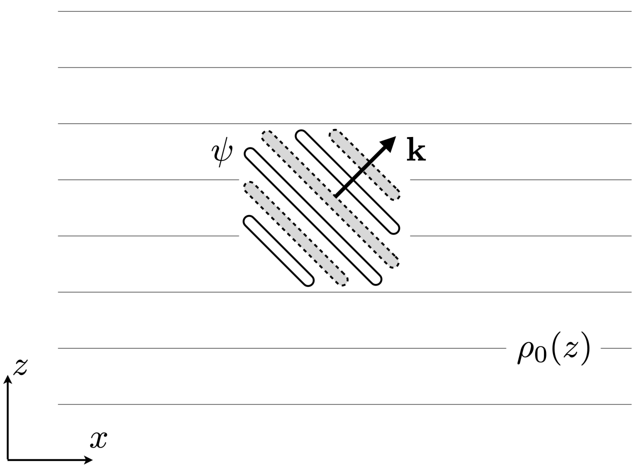
where $b = -g\,\delta\rho/\rho_0$ is the buoyancy anomaly and $\delta\rho$ is the density anomaly relative to $\rho_0(z)$. What is the physical interpretation of this equation? [5]

(c) Define a streamfunction ψ for the flow, stating why this is possible. Write down the linearised equation for the rate of change of buoyancy, and hence show that

$$\frac{\partial^2}{\partial t^2}\left(\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial z^2}\right) = -N^2\frac{\partial^2\psi}{\partial x^2}.$$

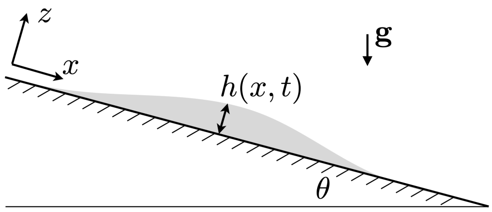
[6]

(d) Obtain the dispersion relation for the resultant wave motion and show that the group velocity is perpendicular to the direction of phase propagation. Sketch how the wave packet shown in the figure evolves and propagates with time.



[10]

4. (a) A thin film of viscous fluid, thickness $h(x,t)$, flows under the influence of gravity down a uniform plane, inclined at an angle θ to the horizontal.



Write down the components of the Navier-Stokes equation parallel and tangential to the plate in the thin film limit, and state the assumptions that lead to these equations. State the appropriate boundary conditions. [7]

(b) Derive an expression for the velocity parallel to the plate. By considering the conservation of mass, or otherwise, show that the thickness of the film evolves according to

$$\frac{\partial h}{\partial t} + \frac{\partial}{\partial x}\left\{\frac{gh^3}{3\nu}\left(\sin\theta - \cos\theta\frac{\partial h}{\partial x}\right)\right\} = 0.$$

[10]

(c) Show that in the limit $\theta = O(1)$, the equation derived in part (b) can be rewritten in the form

$$\frac{\partial h}{\partial t} + u(h,\theta)\frac{\partial h}{\partial x} \approx 0,$$

and state the form of $u(h,\theta)$. Using this result, show that an implicit solution is

$$h \approx h\left(x - \frac{gh^2\sin\theta}{\nu}t\right).$$

Suppose that the thickness of the film is initially as illustrated in the figure in part (a). Sketch, and give a physical explanation for, the subsequent evolution of $h(x,t)$. [8]